

Dental CBCT Tooth Segmentation — Technical Report

Candidate: Manvansh Singh

Assignment: Dobbe AI — Machine Learning Engineer

Date: June 2026

1. Problem Statement

The goal of this assignment was to build an end-to-end machine learning pipeline for dental image segmentation. Specifically, to automatically identify and segment tooth structures from 3D Cone Beam Computed Tomography (CBCT) scans using deep learning.

CBCT scans produce volumetric 3D images of the jaw and teeth. Automated segmentation of these scans has clinical value for treatment planning, implant placement, and orthodontic analysis.

2. Dataset

Dataset: CBCT Teeth Segmentation

Source: Kaggle (`detectioncla/cbct-teeth-segmentation`)

Size: 50 CBCT volumes, ~6.9 GB

Format: NIfTI (.nii), one file per patient

Volume shape: $400 \times 400 \times 435$ voxels

Labels: Per-voxel tooth segmentation masks (29 tooth classes)

This dataset was selected because it is publicly available with no access restrictions, contains genuine 3D CBCT data with expert annotations, and is a manageable size for local training on consumer hardware.

Preprocessing: - Intensity normalization to [0, 1] range - Multi-class labels converted to binary masks (tooth vs background) - Volumes saved as NumPy arrays for fast loading during training

Data split: 34 train / 7 validation / 8 test patients (70/15/15)

3. Model Architecture

A 2D U-Net was selected for this task. U-Net is the standard architecture for medical image segmentation and has been validated extensively on CT and CBCT data.

Approach: The 3D volume is segmented slice-by-slice along the axial axis. Each 2D slice (400 × 400) is passed through the network independently.

Architecture: - Input: single-channel 400 × 400 axial CBCT slice - Encoder: 4 downsampling blocks with feature maps [32, 64, 128, 256] - Each block: Conv2d → BatchNorm → ReLU → Conv2d → BatchNorm → ReLU → MaxPool - Bottleneck: 512 channels - Decoder: 4 upsampling blocks using ConvTranspose2d with skip connections - Output: single-channel binary segmentation mask (sigmoid activation) - Total parameters: 7,762,465

Skip connections between encoder and decoder preserve fine spatial detail, which is critical for accurate tooth boundary delineation.

4. Training

Parameter	Value
Loss function	BCEWithLogitsLoss
Optimizer	Adam
Learning rate	1e-4
LR scheduler	ReduceLROnPlateau (patience=3, factor=0.5)
Batch size	8
Epochs	10
Hardware	NVIDIA RTX 4050 (6GB VRAM)
Training slices	14,790

The model was trained on 2D axial slices extracted from 34 training patients, validated on 7 patients after each epoch, and the best checkpoint was saved based on validation Dice score.

5. Results

Quantitative Results (Test Set — 8 patients, 3,480 slices)

Metric	Score
Mean Dice Score	0.7479
Mean IoU Score	0.6948
Median Dice	0.9104
Best Validation Dice	0.6825

A Dice score of 0.74 is considered good performance for binary tooth segmentation, particularly given the relatively small training set (34 patients) and no data augmentation.

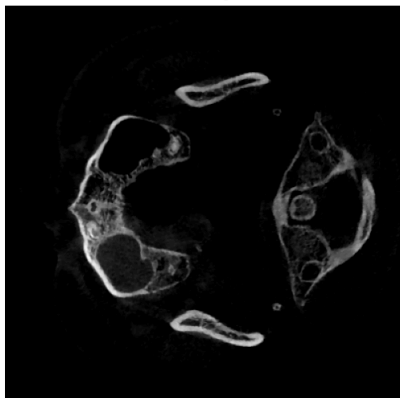
The high median Dice (0.91) reflects strong performance on slices with clear tooth structure. The lower mean is influenced by slices near the top and bottom of the scan where teeth are partially visible and harder to segment.

Qualitative Results

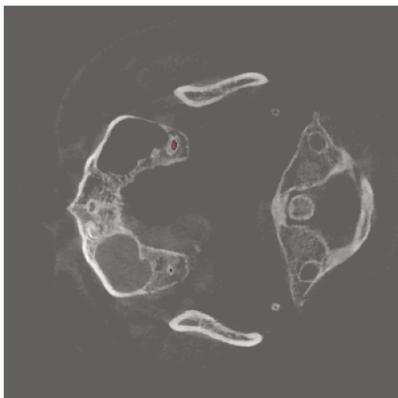
The figure below shows model predictions overlaid on original CBCT slices across 6 representative axial levels. Red indicates ground truth, blue indicates model prediction.

Dental CBCT Segmentation Results

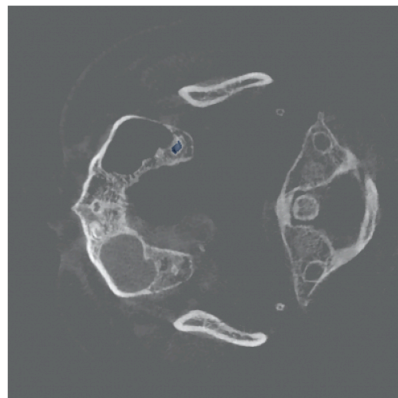
Slice 88 — Original CBCT



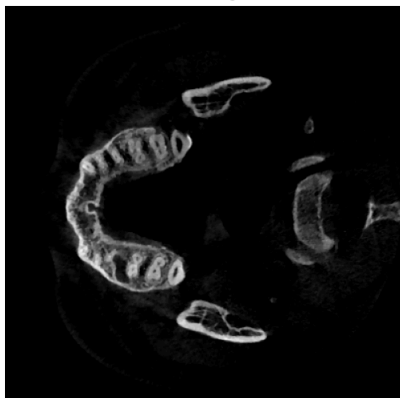
Slice 88 — Ground Truth



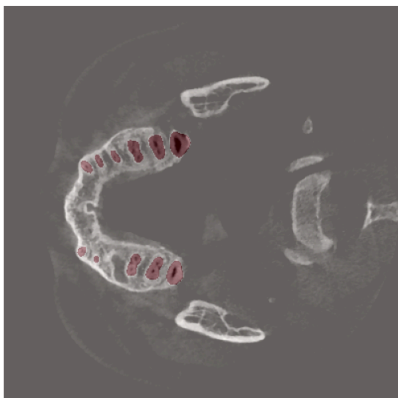
Slice 88 — Model Prediction



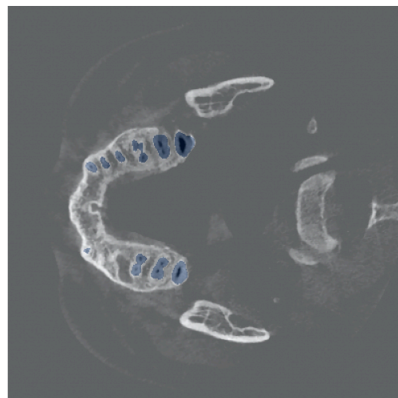
Slice 127 — Original CBCT



Slice 127 — Ground Truth



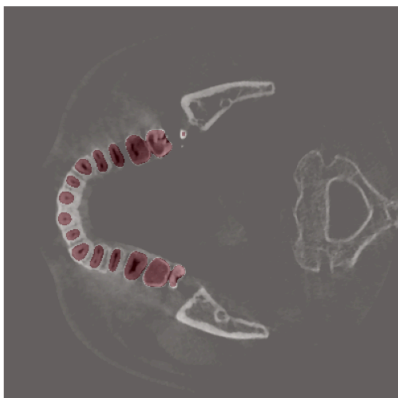
Slice 127 — Model Prediction



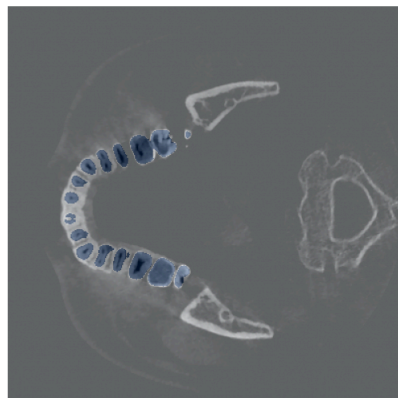
Slice 166 — Original CBCT



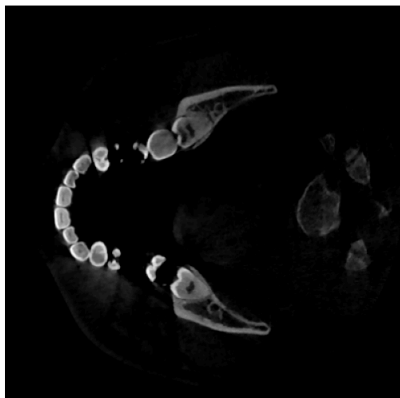
Slice 166 — Ground Truth



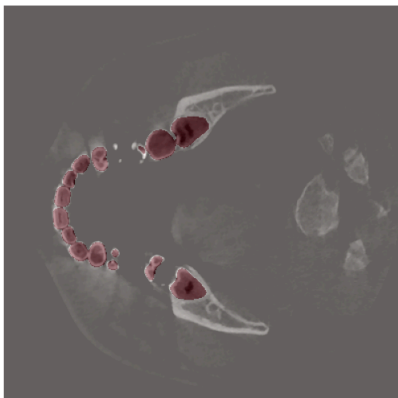
Slice 166 — Model Prediction



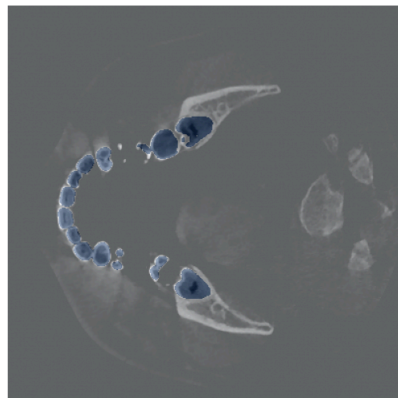
Slice 205 — Original CBCT



Slice 205 — Ground Truth



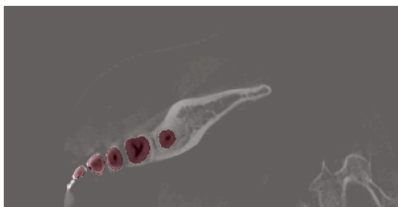
Slice 205 — Model Prediction



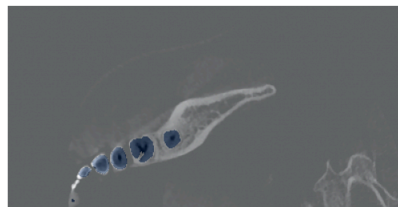
Slice 244 — Original CBCT

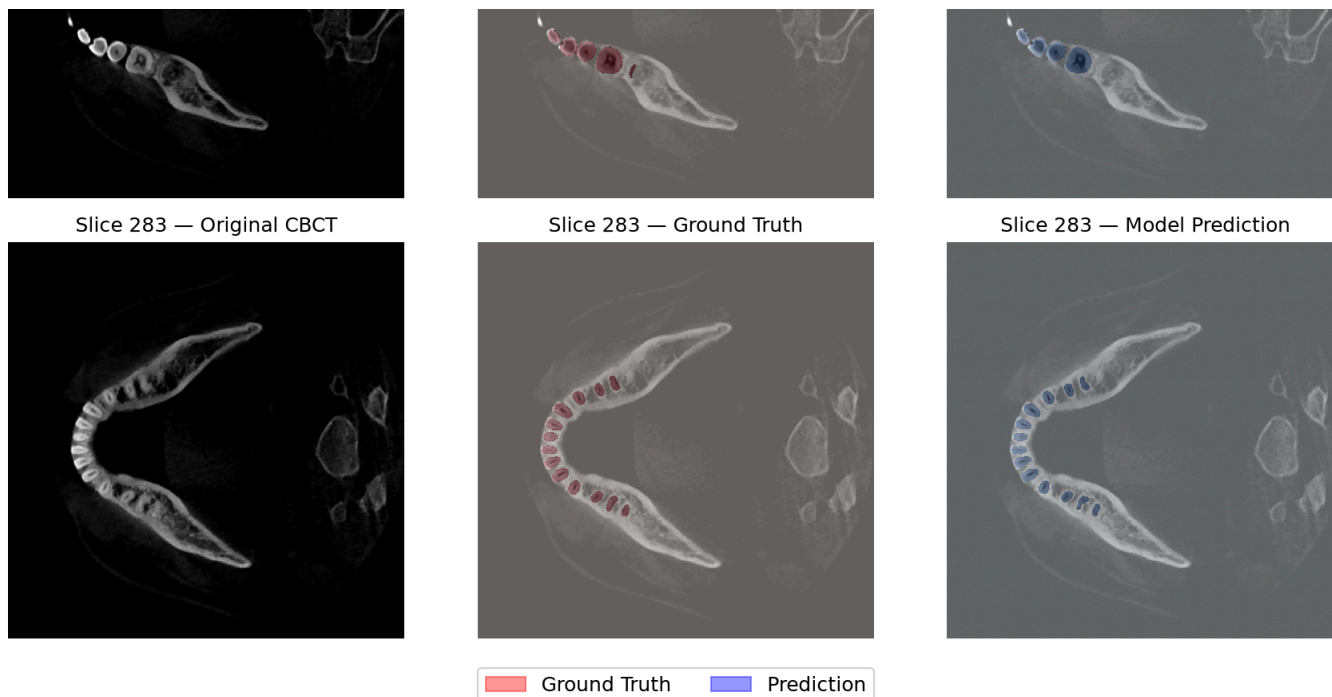


Slice 244 — Ground Truth



Slice 244 — Model Prediction





The model accurately identifies the U-shaped dental arch across all slice levels and correctly localizes individual tooth cross-sections.

An interactive 3D visualization of the full predicted tooth volume overlaid on the CBCT scan is provided as [outputs/figures/3d_visualization.html](#).

6. Post-processing

After inference, small isolated voxel clusters (fewer than 500 voxels) are removed using connected component analysis, and holes within predicted tooth regions are filled using binary morphological operations. This removed 757 noise voxels (0.1% of predictions) on the test patient, indicating the model produces clean predictions with minimal spurious activations.

7. Limitations and Future Work

- **2D vs 3D:** Slice-by-slice inference does not use inter-slice context. A 3D U-Net or sliding window approach would likely improve boundary consistency.
- **Binary segmentation:** The pipeline predicts tooth vs background. Multi-class per-tooth segmentation would enable individual tooth tracking.
- **Dataset size:** 34 training patients is small for medical imaging. More data or transfer learning from a pretrained encoder would improve generalization.
- **Augmentation:** No data augmentation was applied. Adding rotation, flipping, and intensity jitter would reduce overfitting.
- **Evaluation:** Hausdorff distance and surface Dice would give a more complete picture of boundary accuracy.

8. Pipeline Summary

Raw NIfTI → Normalize → Extract 2D slices → U-Net → Binary mask → Postprocess →
Predicted NIfTI + 3D Visualization

All code is modular and documented. Running the full pipeline from a newscan requires a single call to `inference.py` .